

THE DEGREE-SIX TERNARY KELLER THEOREM: FIRST-DISAPPEARANCE GEOMETRY AND POINTED COMPANION EXCLUSION

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ABSTRACT. Let

$$F : \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3, \quad \det JF \in \mathbb{C}^{\times}, \quad \max_i \deg F_i \leq 6.$$

We prove that F is a polynomial automorphism. A hypothetical nonproper map produces a ramified finite dicritical divisor with negative common-degree-six defect. A tuned relative minimal-model program carries this divisor to its first divisorial disappearance while the original ordinary-degree data are retained in a finite unsaturated Rees diagram. At every irreducible branch of the resulting special fibre, a componentwise presentation gives either a regular determinant-balanced package or a pointed quadratic or cubic companion over the algebra of all remaining simultaneous keys. Regular curve branches are excluded by a filtered inverse argument, while regular point branches reduce to six hyperbolic signatures and forty explicitly described locally closed coefficient strata. For singular branches, denominator-free quadratic and cubic symmetrizers produce universal terminal Euler–Jacobian rows satisfying

$$Q_0 - \gamma Q_1 = 2, \quad \gamma^2 E_2 - (\gamma\tau - 1)E_0 - 2\gamma E_1 = 3.$$

A primewise polynomial base change sends these rows into the actual localized Keller ideal. Hence no companion component occurs, and the selected dicritical divisor has no possible first disappearance.

All structural arguments are given in the body of the paper. The remaining finite algebra consists of two explicit eliminations: a twenty-four-equation Pluecker system for the direct $(1, 2, 3)$ model, and the forty regular hyperbolic coefficient ideals. Their defining equations and normalizations are recorded in the appendices, so the finite algebraic input is part of the manuscript itself.

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1. INTRODUCTION AND STATEMENT OF THE THEOREM

A polynomial map is called a *Keller map* if its Jacobian determinant is a nonzero constant.

Theorem 1.1 (Degree-six ternary Keller theorem). *Every Keller map*

$$F : \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3, \quad \max_i \deg F_i \leq 6,$$

is a polynomial automorphism.

The proof is organized around the boundary of the projective extension of F . Its main steps are as follows.

- (i) Nonproperness produces a ramified finite dicritical divisor E with a negative common-degree-six defect. A relative klt minimal-model program must therefore contract E .
- (ii) The valuation v_E defines a finite Rees diagram containing the original ordinary-degree source spaces, all derivative and mixed-partial graphs, the Euler first column, every determinant product, the raw Pluecker minors, and all key tails needed through degree six. This diagram is stationary before the first disappearance of E .
- (iii) At the generic point of every irreducible first-disappearance branch, visible source relations are eliminated componentwise. The branch is either regular or contains a pointed rank-two or rank-three companion over the algebra generated by all other simultaneous keys.
- (iv) Regular point and curve branches are excluded separately. Pointed companions are excluded by universal terminal coefficient identities and a primewise base-change theorem.

The purpose of the direct calculations is limited and explicit. The direct $(1, 2, 3)$ model is reduced to the polynomial system written in Section A; the direct $(1, 1, 4)$ model supplies the standard rank-three companion algebra and the three terminal coefficient rows. The regular point calculation is organized by the orbit stratification in Section B. No other finite coefficient computation enters the proof.

2. COMMON-DEGREE-SIX GEOMETRY AND FIRST DISAPPEARANCE

Put every target component in common projective degree six:

$$G_0 = X_0^6, \quad G_i = X_0^{6-\deg F_i} F_i^h \quad (1 \leq i \leq 3). \quad (1)$$

Lemma 2.1 (Homogeneous Jacobian identity). *If $\det JF = \kappa \in \mathbb{C}^\times$, then*

$$\det J(G_0, G_1, G_2, G_3) = 6\kappa X_0^{20}. \quad (2)$$

Proof. Each row of the homogeneous Jacobian has degree five, so the determinant has degree twenty. On the affine chart $X_0 = 1$, expansion along the first row gives $6 \det JF = 6\kappa$. The only homogeneous polynomial of degree twenty whose restriction to $X_0 = 1$ is the constant 6κ is $6\kappa X_0^{20}$. $\square$

Let

$$\pi : X \longrightarrow \mathbb{P}^3, \quad \phi : X \longrightarrow \mathbb{P}^3$$

be a smooth resolution of the projective graph. For a finite dicritical prime divisor E , set

$$m_E = v_E(\pi^* X_0), \quad q_E = \min_i v_E(\pi^* G_i),$$

and let k_E and r_E denote the coefficients of E in $K_X - \pi^* K_{\mathbb{P}^3}$ and $K_X - \phi^* K_{\mathbb{P}^3}$, respectively. Define

$$\delta_E = q_E - k_E - 2m_E.$$

Proposition 2.2 (Sharp common-degree-six defect). *If F is nonproper, there is a ramified finite dicritical divisor E such that*

$$q_E = 6m_E, \quad r_E = k_E - 4m_E > 0, \quad \boxed{\delta_E = -r_E < 0}.$$

Proof. Zariski's Main Theorem factors the quasi-finite étale map through the normalization of the affine target in the source function field. If the finite normalization were unramified in codimension one, purity would make it finite étale; affine space has no nontrivial connected finite étale cover. Thus a ramified boundary divisor mapping to the affine target exists.

At its generic point an affine target coordinate is nonzero. After removal of the common base order this gives $q_E = v_E(G_0) = 6m_E$. Comparing source and target volume forms gives $r_E = k_E - 4m_E$, and ramification gives $r_E > 0$. Substitution proves the formula for δ_E . $\square$

Let $a = 1/3$ and $\lambda = 2/3$. For two general members D_1, D_2 of the complete projective Keller system, with strict transforms S_1, S_2 , the coefficient of E in

$$K_X + a(S_1 + S_2) - \pi^*(K_{\mathbb{P}^3} + a(D_1 + D_2))$$

is

$$k_E - \lambda q_E = -\delta_E > 0.$$

Run the corresponding relative klt MMP and stop when E disappears.

Lemma 2.3 (First disappearance). *Immediately before its first disappearance, E is the exceptional divisor of a terminal divisorial contraction*

$$f : X' \longrightarrow Y$$

whose centre is a point or a curve. At the generic point of a curve centre, f is the ordinary blow-up of the smooth generic centre.

Proof. Existence and termination of the threefold relative MMP, together with the negativity lemma, force the positive exceptional coefficient to disappear. A flip is small, so the first step removing E is divisorial. The generic description of a terminal divisorial contraction to a curve is the ordinary blow-up; point contractions are covered by the standard threefold classification. $\square$

3. THE STATIONARY FINITE ORDINARY-DEGREE DIAGRAM

Let $K = \mathbb{C}(x_1, x_2, x_3)$ and $v = v_E$. For a finite-dimensional polynomial space $W \subset K$, set

$$F^q W = \{w \in W : v(w) \geq q\}, \quad \mathcal{R}_v(W) = \sum_q F^q W t^{-q}.$$

Lemma 3.1 (Finite Rees freeness). *There is a filtration-adapted basis e_i with jumps λ_i such that*

$$\mathcal{R}_v(W) = \bigoplus_i \mathbb{C}[t] t^{-\lambda_i} e_i, \quad \mathcal{R}_v(W)/t\mathcal{R}_v(W) \simeq \mathrm{gr}_v W$$

canonically.

Proof. Choose lifts of bases of the finitely many nonzero graded quotients. The decomposition and the special-fibre identification follow immediately. $\square$

Definition 3.2 (Stationary finite diagram). The diagram $\mathcal{K}_v^{\leq 6}$ contains the Rees modules of

- (a) all source spaces of ordinary degree at most six and the complete target degree flag;
- (b) all derivative and mixed-partial graphs;
- (c) the Euler first column $6F_i - \mathcal{E}F_i$;
- (d) every tensor-product space entering (2), through ordinary degree twenty;
- (e) the raw Pluecker minors before saturation;
- (f) every normal-subduction key tail through ordinary cost six.

A multilinear map is first viewed as a linear map from the appropriate tensor product and is then retained through its graph.

Proposition 3.3 (Stationarity). *The diagram $\mathcal{K}_v^{\leq 6}$ is finite and free over $\mathbb{C}[t]$. Every polynomial identity in the diagram survives specialization to $t = 0$. Before first disappearance, every birational step gives the same diagram at the generic point of E , up to multiplying t by a unit.*

Proof. Apply Theorem 3.1 to every object in Theorem 3.2. Retaining a map by its graph preserves polynomial identities without requiring kernels or images to commute with specialization. Before first disappearance the birational models are isomorphic at the generic point of the strict transform of E , so the valuation ring and the original ordinary-degree spaces are unchanged. $\square$

A constant target basis adapted to

$$L_{\leq e} = \text{Span}_{\mathbb{C}}\{1, F_1, F_2, F_3\} \cap \mathbb{C}[x_1, x_2, x_3]_{\leq e}$$

places every lower actual target degree in the common-degree-six system with its forced power of X_0 . Thus deficient top-degree spans are closed linear strata of the same diagram.

Proposition 3.4 (Finite-window sufficiency). *The componentwise argument requires no presentation of the full associated-graded source algebra. A relation of ordinary cost greater than six can affect the proof only through a retained tensor, determinant, or raw-minor object; in that case the induced linear dependence is already encoded in the Rees module of the actual finite-dimensional object.*

Proof. The proof uses a finite diagram of actual polynomial subspaces, not the free tensor algebra on abstract source symbols. Every product space entering the homogeneous determinant or a raw Pluecker coefficient is inserted as its actual finite-dimensional image before specialization. Hence every dependence which occurs in an equation used later is already present in the corresponding Rees module. A higher-cost relation with no image in any retained object cannot alter a later Keller equation. $\square$

4. RELATIVE NORMAL PACKAGES AND POINTED COMPANIONS

Let $Z = f(E)$ and put

$$k_Z = \begin{cases} \mathbb{C}, & \dim Z = 0, \\ \mathbb{C}(Z), & \dim Z = 1. \end{cases}$$

The special fibre of $\mathcal{K}_v^{\leq 6}$ over k_Z is the *relative normal package*. It retains the original ordinary costs even though the local analytic coordinates on Y need not be original source coordinates.

Lemma 4.1 (Separating coordinate and vertical determinant). *Assume Z is a curve. After a finite separable extension of k_Z and a constant target change, there is a zero-weight target coordinate S with $S_s \in k_Z^\times$. Define*

$$\partial_u^S = \partial_u - \frac{S_u}{S_s} \partial_s, \quad \partial_v^S = \partial_v - \frac{S_v}{S_s} \partial_s.$$

Then

$$\det \frac{\partial(S, A, B)}{\partial(s, u, v)} = S_s \det \begin{pmatrix} \partial_u^S A & \partial_v^S A \\ \partial_u^S B & \partial_v^S B \end{pmatrix}. \quad (3)$$

Proof. Finite dicriticality and characteristic zero provide a separating residue coordinate after a finite separable base change. The two displayed vertical derivatives are obtained from ∂_u, ∂_v by column operations which kill the S -entries. The determinant is then block triangular. $\square$

Choose the finitely many source and key symbols of ordinary cost at most six and write

$$\mathcal{A}_{\leq 6} = k_Z[Z_1, \dots, Z_N]/J_{\leq 6}.$$

A relation is *visible* if its ordinary cost is at most six. Linear elimination is performed only where its coefficient is a unit; every coefficient-zero divisor is retained by Noetherian induction.

Lemma 4.2 (Complete nonlinear cost list). *Let a key W have cost $j \geq 2$. If a primitive visible relation has exact degree $n \geq 2$ in W and total cost c , then $nj \leq c \leq 6$. Therefore*

$$(j, n, c) = (2, 2, 4), (2, 2, 5), (2, 2, 6), (2, 3, 6), (3, 2, 6). \quad (4)$$

In particular, the nonlinear companion rank is two or three.

Proof. The monomial W^n has cost at least nj . If $j, n \geq 2$ and $nj \leq 6$, then $(j, n) = (2, 2), (2, 3)$ or $(3, 2)$. Allowing all c with $nj \leq c \leq 6$ gives exactly (4). $\square$

Lemma 4.3 (Pointed componentwise reduction). *Let η be the generic point of an irreducible branch of $\text{Spec } \mathcal{A}_{\leq 6}$. After shrinking and normalizing the branch, a primitive visible relation either eliminates a generator linearly or yields a finite locally free pointed algebra*

$$\mathcal{B}_n = \frac{\mathcal{C}[W]}{(W^n + a_1 W^{n-1} + \dots + a_{n-1} W)}, \quad n = 2 \text{ or } 3, \quad (5)$$

where $\mathcal{C}$ contains every other simultaneous key, coefficient parameter, and retained graph output. The map $W \mapsto 0$ is a canonical augmentation.

Proof. Choose a visible relation of minimal cost after the available linear eliminations. If it is linear in a remaining generator, eliminate that generator on the open set where its coefficient is a unit and treat the coefficient-zero components separately.

Otherwise choose a key W occurring nonlinearly. In the normal-subduction presentation the next key is the remainder of the primitive relation. Moving that remainder to the right and restricting to its first singular boundary puts the relation in the form

$$W^n + a_1 W^{n-1} + \dots + a_{n-1} W = 0.$$

After content division, primitive factorization, and generic freeness, its nonlinear rank is two or three by Theorem 4.2. Monic division gives the basis $1, W, \dots, W^{n-1}$. All other simultaneous keys remain in $\mathcal{C}$; they need not be linear or algebraically independent. $\square$

Theorem 4.4 (Componentwise first-disappearance alternative). *At the generic point of every irreducible first-disappearance branch, exactly one of the following holds for the finite stationary diagram.*

- (R1) *The source-coordinate window is regular and the first nonzero determinant layer is an honest determinant-balanced regular package.*
- (R2) *There is a pointed companion (5) of rank two or three. Every other simultaneous key remains in $\mathcal{C}$, and the selected companion direction, together with multiplication, mixed partials, the Euler column, determinant products, and raw Pluecker maps, is a polynomial base change of the universal pointed rank- n package.*

The alternatives commute with passage to every coefficient-divisor branch.

Proof. Apply Theorem 4.3 recursively to the finite list of visible relations. If all steps are linear, the surviving source-coordinate window is free and the retained determinant graph gives (R1); higher product dependences remain encoded by Theorem 3.4.

Otherwise stop at the first pointed monic factor. Monic division gives the basis $1, W, \dots, W^{n-1}$. Multiplication, graph outputs, mixed partials, the Euler column, determinant products, and raw minors are polynomial in the structure coefficients and retained coefficient variables. Thus they arise from the universal pointed package by base change. A second relation involving W reduces to degree $< n$; at the generic prime it lowers the selected rank or vanishes, and after shrinking it is absorbed into the coefficient algebra. No simultaneous relation or coefficient divisor is discarded. $\square$

5. REGULAR RELATIVE ENDPOINTS

5.1. Regular point normal form and the master equation.

Lemma 5.1 (Regular point normal form). *For a non-plane regular point package, after primitive regrading and a homogeneous change of generators, the source weights are*

$$(1, -r, -s), \quad 1 \leq r \leq s,$$

and the determinant-balanced target shifts are $(-s, -r, 1)$.

Proof. Choose the primitive positive Rees generator x . The remaining homogeneous generators have the form $u = x^r y$ and $v = x^s z$, with $0 \leq r \leq s$; finite dicriticality rules out $r = 0$ except for a free plane factor. Primitive regrading and determinant balance give the stated weights and target shifts. $\square$

Put $u = x^r y$ and $v = x^s z$. The descended map is

$$G = (x^{-s}A(u, v), x^{-r}B(u, v), x\Lambda(u, v)),$$

and the determinant equation is

$$\Lambda\{B, A\} + sA\{B, \Lambda\} + rB\{\Lambda, A\} = 1, \quad \{P, Q\} = P_u Q_v - P_v Q_u. \quad (6)$$

The ordinary-degree supports are

$$\begin{aligned} S_A &= \{(i, j) : ri + sj \geq s, (r+1)i + (s+1)j - s \leq 6\}, \\ S_B &= \{(i, j) : ri + sj \geq r, (r+1)i + (s+1)j - r \leq 6\}, \\ S_\Lambda &= \{(i, j) : 1 + (r+1)i + (s+1)j \leq 6\}. \end{aligned} \quad (7)$$

5.2. Uniform theoretical exclusions.

Lemma 5.2 (Uniform exclusions). *If Λ is nonconstant, then after the uniform coefficient comparisons only*

$$(r, s) = (1, 1), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4)$$

remain.

Proof. The support condition on S_Λ first leaves

$$(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4).$$

For $s \geq 5$ one has $A = vC(u) + D(u)$, $B = B_0(u) + bv$, and $\Lambda = L(u)$. The coefficient of v in (6) is $-b(LC' + (s-r)L'C)$.

If $b \neq 0$, comparing the multiplicity of every irreducible factor of L and C forces L to be constant. If $b = 0$, the constant term is $C(LB'_0 + rB_0L')$, and the same factor-multiplicity argument again makes L constant.

For $r = s = q \geq 2$, the quadratic coefficient operator induced by (6) has eigenvalues $-q, 1, q+2$ on u^2, uv, v^2 . None vanishes, so the quadratic part of Λ is zero; the linear comparison then forces Λ to be constant. Finally, the $(3, 4)$ support reduces to variables X, Y satisfying

$$6X + 5Y = 0, \quad 2X + 2Y + 4 = 0, \quad X = 0,$$

a contradiction. The six displayed signatures remain. $\square$

5.3. Normal forms and the residual finite ideals. The constant term of Λ is normalized to 1. When $r < s$, the constant Jacobian at the origin permits constant source and target changes making the linear parts of A, B equal to v, u , respectively. The residual grading torus acts diagonally on the coefficients of $\Lambda - 1$.

Proposition 5.3 (Exhaustive normal forms). *Every nonconstant- Λ solution of (6) in the six signatures of Theorem 5.2 is equivalent, under the allowed constant changes and grading tori, to one of the forty locally closed normal forms in Section B. They are grouped as*

$$2 + 2 + 3 + 3 + 5 + 25 = 40$$

forms for the signatures (2, 4), (2, 3), (1, 3), (1, 4), (1, 1), (1, 2), respectively.

Proof. For (2, 3) and (2, 4), (7) gives $\Lambda - 1 = \ell_{10}u + \ell_{01}v$; diagonal scaling gives the pure- u and v -present orbits. For (1, 3) and (1, 4) one has $\Lambda - 1 = \ell_{10}u + \ell_{01}v + \ell_{20}u^2$, giving the three stated orbits.

For (1, 1), the quadratic part of $\Lambda - 1$ is a binary quadratic form. The GL_2 action classifies it by rank 0, 1, 2. In rank one the linear part is tangent or transverse to the doubled line; in rank two it is zero or nonzero. This gives five orbits.

For (1, 2),

$$\Lambda - 1 = \ell_{10}u + \ell_{01}v + \ell_{20}u^2 + \ell_{11}uv.$$

The grading torus gives the five non-pure families recorded in Section B. Within each family, coefficient comparison in (6) orders the remaining cases by the first nonzero member of $a_{40}, a_{12}, a_{21}, a_{30}$, with a final closed stratum. At each open stage the named coefficient is normalized; its zero locus is retained as the next stage. The resulting decision tree has twenty-five leaves. $\square$

Remark 5.4 (Structure of the residual point calculation). The forty strata are not forty unrelated computations. They arise from four uniform orbit mechanisms.

For (2, 3) and (2, 4), the support of $\Lambda - 1$ is linear, and the only invariant distinction is whether the v -coefficient vanishes. For (1, 3) and (1, 4), the possible quadratic term is a multiple of u^2 , so the strata are ordered by the first nonzero coefficient among $\ell_{01}, \ell_{20}, \ell_{10}$. For (1, 1), the binary quadratic part is classified by rank, and the linear part is measured relative to its null line. Only the signature (1, 2) has a genuine coefficient tree: after fixing the torus type of $\Lambda - 1$, the master equation forces certain b -coefficients to vanish and leaves the successive pivots

$$a_{40}, \quad a_{12}, \quad a_{21}, \quad a_{30}.$$

The open pivot is normalized, while its zero locus is retained as the next closed stage. Thus the finite elimination is compatible with specialization and does not discard a boundary stratum.

For a normal form ν , let $A_\nu, B_\nu, \Lambda_\nu$ be the generic polynomials with supports (7) after imposing the normalizations in Section B. Let I_ν be the ideal generated by the coefficients of

$$\Lambda_\nu\{B_\nu, A_\nu\} + sA_\nu\{B_\nu, \Lambda_\nu\} + rB_\nu\{\Lambda_\nu, A_\nu\} - 1,$$

together with a relation $z_f f - 1$ for every coefficient f required to be nonzero.

Proposition 5.5 (Finite regular elimination). *For every one of the forty normal forms ν in Section B,*

$$I_\nu = (1).$$

Proof. The ideals are completely determined by (7), (6), and the table in Section B. Expanding the master equation and eliminating the coefficient variables in graded reverse lexicographic order yields the reduced basis $\{1\}$ in each case. No open condition is implemented by division: every such condition is represented by a Rabinowitsch equation, so the complementary closed branch is one of the subsequent normal forms. The numbers of variables and equations for all forty ideals are included in the table to make the finite calculation directly reconstructible. $\square$

Theorem 5.6 (Regular point endpoint). *A regular point-centred first-disappearance package of ordinary degree at most six is an automorphism or is empty.*

Proof. If Λ is nonconstant, use Theorems 5.2, 5.3 and 5.5. If Λ is constant, (6) is a plane Keller equation in the indicated low-degree supports; the classical plane low-degree theorem makes the normal pair a polynomial automorphism. $\square$

5.4. Regular curve endpoints.

Lemma 5.7 (Filtered inverse criterion). *Let a morphism of complete filtered local algebras over a characteristic-zero field preserve the maximal-ideal filtration. If its associated-graded map is a polynomial automorphism and its residue-field map is separable, then the completed morphism is an isomorphism. In particular, the induced extension of divisorial valuation rings is unramified.*

Proof. Lift the inverse degree by degree. At each stage the error in the next graded piece is removed by the inverse graded linear map. Completeness gives mutually inverse continuous homomorphisms. Separability identifies the completed residue extension, and an isomorphism of complete valuation rings has different exponent zero. $\square$

Theorem 5.8 (Regular curve endpoint). *A regular curve-centred first-disappearance package of ordinary degree at most six is impossible for a ramified finite dicritical divisor.*

Proof. By Theorem 4.1, the two normal source generators form a polynomial algebra over k_Z and their vertical Jacobian is a unit. Their ordinary costs are at most six. After algebraic closure, the plane low-degree theorem makes the normal pair a polynomial automorphism; uniqueness descends the inverse to k_Z . Together with the separating residue coordinate, the associated-graded map is an isomorphism. By Theorem 5.7, the completed map is an isomorphism, so the different exponent is zero, contrary to $r_E > 0$. $\square$

6. THE DIRECT $(1, 2, 3)$ MODEL

Put $w = y - h^2$ and give (h, x, y) weights $(1, 2, 3)$.

Lemma 6.1 (The rank-68 bounded module). *The degree-at-most-six source space in valuation weight at least six has dimension 68.*

Proof. The full polynomial space has dimension $\binom{9}{3} = 84$. The monomials of weight below six are

$$\begin{array}{l|l} 0 & 1 \\ 1 & h \\ 2 & h^2, x \\ 3 & h^3, hx, y \\ 4 & h^4, h^2x, x^2, hy \\ 5 & h^5, h^3x, hx^2, h^2y, xy. \end{array}$$

There are sixteen, hence the required dimension is $84 - 16 = 68$. $\square$

Define

$$\mathcal{D} = h\partial_h + 2x\partial_x + (3y - h^2)\partial_y - 6$$

and

$$\mathcal{W}(f, g, k) = \det \begin{pmatrix} \mathcal{D}f & f_x & f_y \\ \mathcal{D}g & g_x & g_y \\ \mathcal{D}k & k_x & k_y \end{pmatrix}.$$

Let

$$\begin{aligned} f_0 &= h^3y - hy^2, & f_1 &= h^3x - hxy, & f_2 &= h^5 - hy^2, \\ V_7 &= \text{Span}_{\mathbb{Q}}\{hy^3, hxy^2, hx^2y, hx^3, h^3y^2, h^3xy, h^3x^2, h^5y, h^5x\}. \end{aligned}$$

Write $F = Af_0 + Bf_1 + f_2$, choose the displayed ordered basis $v_0, \dots, v_8$ of V_7 , and write

$$p = \sum_{0 \leq i < j \leq 8} p_{ij} v_i \wedge v_j.$$

The Keller equation in this chamber is

$$\mathcal{W}(F, p) = h^{15}.$$

Proposition 6.2 (Polynomial compatibility syzygy). *Let I_W be the coefficient ideal of $\mathcal{W}(F, p) - h^{15}$. Then*

$$B^5 \in I_W.$$

Consequently B belongs to every prime containing I_W .

Proof. Let C_μ be the coefficient row indexed by the monomial $h^{\mu_0} x^{\mu_1} y^{\mu_2}$. The multipliers λ_μ displayed in Section A satisfy

$$\sum_{\mu} \lambda_{\mu} [p_{ij}] C_{\mu} = 0 \quad \text{for every } p_{ij}, \quad \sum_{\mu} \lambda_{\mu} C_{\mu} = -B^5.$$

Both identities follow by direct substitution into the determinant defining $\mathcal{W}$. Thus B^5 is a polynomial combination of the coefficient rows. $\square$

Modulo B , the coefficient of h^{15} is $p_{78} - 1$. Hence the closed fibre lies in the Pluecker big cell $p_{78} = 1$. Every decomposable two-vector in this cell has unique coordinates $a_0, \dots, a_6, b_0, \dots, b_6$ with

$$p_{i7} = a_i, \quad p_{i8} = b_i, \quad p_{ij} = a_i b_j - b_i a_j \quad (0 \leq i < j \leq 6). \quad (8)$$

Proposition 6.3 (Closed-fibre Pluecker elimination). *After substituting $B = 0$ and (8), the remaining coefficient equations are the twenty-four polynomials $R_0, \dots, R_{23}$ listed in Section A. They generate the unit ideal in*

$$\mathbb{Q}[A, a_0, \dots, a_6, b_0, \dots, b_6].$$

Proof. The first five equations successively control the last Pluecker coordinates, and the remaining equations impose the lower exterior compatibilities. Exact polynomial elimination in the displayed variable ring reduces the ideal to $\{1\}$. Since the complete list is written in Section A, the finite calculation is determined entirely by the displayed equations. $\square$

Lemma 6.4 (Nilpotent closed-fibre criterion). *Let R be a ring, $I \subset R$ an ideal, and $b \in R$. If $b^N \in I$ for some $N \geq 1$ and $I + (b) = R$, then $I = R$.*

Proof. In R/I , the class of b is both nilpotent and a unit. Hence $1 = 0$ in R/I . $\square$

Theorem 6.5 (Direct (1, 2, 3) exclusion). *The direct (1, 2, 3) model has no degree-at-most-six Keller solution.*

Proof. By Theorem 6.2, the full coefficient ideal contains B^5 . By Theorem 6.3, its quotient after adjoining B is the unit ideal. Apply Theorem 6.4. $\square$

Remark 6.6 (Why the closed fibre is separate). A left-kernel computation over the function field $\mathbb{Q}(A, B)$ does not control specialization to $B = 0$, because the coefficient matrix drops rank on that divisor. The polynomial syzygy and the big-cell argument above retain the rank-jump fibre scheme-theoretically.

7. UNIVERSAL POINTED COMPANION ALGEBRAS

7.1. The quadratic algebra. For a quadratic core over a local characteristic-zero algebra A , put

$$B_2 = A[b]/(b^2 + \alpha b), \quad \epsilon(b) = 0, \quad I_2 = Ab. \quad (9)$$

A graph correction has the form $h(b) = c + mb$. Modulo the lower translation slot, its reduced coordinate is

$$\rho_2([h]) = m \in \text{End}_A(I_2) \simeq A. \quad (10)$$

7.2. The rank-three determinantal algebra. Let U, z, s, a, c have ordinary costs 1, 1, 2, 3, 4, and let $\alpha(U), \beta(U)$ have degrees at most 2, 3. Consider

$$B = \frac{\mathbb{Q}[U, z, s, a, c]}{I_2 \begin{pmatrix} s & a & c \\ z & s - \alpha(U) & a - \beta(U) \end{pmatrix}}. \quad (11)$$

The three relations are

$$s(s - \alpha) = az, \quad s(a - \beta) = cz, \quad a(a - \beta) = c(s - \alpha). \quad (12)$$

Proposition 7.1 (Standard monomial form and rank 60). *Every class in the ordinary-degree-at-most-six part of B has a unique form*

$$P(U, z, c) + sQ(U, z, c) + aT(U, z, c), \quad (13)$$

with

$$\deg P \leq 6, \quad \deg Q \leq 4, \quad \deg T \leq 3.$$

This space has dimension

$$34 + 16 + 10 = 60.$$

Proof. With a term order in which s^2, sa, a^2 are leading, the three relations in (12) form a Groebner basis. Their two Hilbert–Burch syzygies are

$$s\mu_6 - a\mu_5 + c\mu_4 = 0, \quad z\mu_6 - (s - \alpha)\mu_5 + (a - \beta)\mu_4 = 0,$$

where μ_4, μ_5, μ_6 are the three minors. The standard monomials contain at most one factor s or one factor a , giving (13) and uniqueness.

For P , the monomials without c contribute $\binom{8}{2} = 28$ and those with one c contribute $\binom{4}{2} = 6$, hence 34. For Q , the corresponding count is $\binom{6}{2} + 1 = 16$. For T , the variable c cannot occur and the count is $\binom{5}{2} = 10$. $\square$

Corollary 7.2 (Hilbert series of the determinantal algebra). *Let $S = \mathbb{Q}[U, z, s, a, c]$ with weights 1, 1, 2, 3, 4. Then*

$$H_B(t) = \frac{1 - t^4 - t^5 - t^6 + t^7 + t^8}{(1 - t)^2(1 - t^2)(1 - t^3)(1 - t^4)}.$$

Proof. The three minors have weights 4, 5, 6. The two Hilbert–Burch syzygies in the proof of Theorem 7.1 have weights 7 and 8, giving the weighted free resolution

$$0 \longrightarrow S(-7) \oplus S(-8) \longrightarrow S(-4) \oplus S(-5) \oplus S(-6) \longrightarrow S \longrightarrow B \longrightarrow 0.$$

Taking Hilbert series proves the formula. Its coefficients through weight six agree with the standard-monomial count $34 + 16 + 10 = 60$. $\square$

Lemma 7.3 (Denominator-free cubic normal form). *Let A be a DVR and let $B = A \oplus I$ be a finite free augmented commutative algebra of rank three. Assume that generically $B_K \simeq K \times L$, with L/K quadratic and the augmentation equal to the projection onto K . Then I has a basis u, v such that*

$$u^2 = -\alpha u, \quad uv = -\alpha v, \quad v^2 = cu - \beta v \quad (14)$$

for some $\alpha, \beta, c \in A$. Equivalently,

$$B \simeq \frac{A[u, v]}{I_2 \begin{pmatrix} v & u & 0 \\ c & v + \beta & u + \alpha \end{pmatrix}}.$$

No structure coefficient is inverted.

Proof. The generic augmentation ideal $I_K \simeq L$ has an identity element e . Choose $\alpha \in A$ of minimal valuation such that $u = -\alpha e \in I$. Then u is primitive and extends to a basis u, v of I . Multiplication by u on I is $-\alpha$ times the identity, so $u^2 = -\alpha u$ and $uv = -\alpha v$. Write $v^2 = cu - \beta v$. The three relations are precisely the 2×2 minors of the displayed matrix. $\square$

7.3. The cubic symmetrizer. For an augmented algebra set

$$\text{Symm}(B, \epsilon) = \{h : I \rightarrow B : xh(y) = yh(x) \text{ for all } x, y \in I\}.$$

Write

$$h(u) = A_0 + pu + rv, \quad h(v) = B_0 + qu + sv.$$

The equation $uh(v) = vh(u)$ gives

$$A_0 = \alpha p + \beta r - \alpha s, \quad B_0 = \alpha q + cr. \quad (15)$$

Hence projection to I is a denominator-free isomorphism

$$\text{Symm}(B, \epsilon) \xrightarrow{\sim} \text{End}_A(I), \quad h \mapsto M = \begin{pmatrix} p & q \\ r & s \end{pmatrix}. \quad (16)$$

Thus the cubic symbol space has rank four. Its exterior ranks are

$$\binom{4}{1}, \binom{4}{2}, \binom{4}{3}, \binom{4}{4} = 4, 6, 4, 1.$$

7.4. Universal terminal coefficient rows. Let γ be the final scalar target shear. In rank two, the reduced graph parameter is m , and the last two Euler–determinant coefficients are

$$Q_1 = m, \quad Q_0 = 2 + \gamma m. \quad (17)$$

Therefore

$$Q_0 - \gamma Q_1 = 2. \quad (18)$$

In rank three put

$$\tau = \text{Tr } M, \quad \sigma = \det M.$$

In an adapted order-four frame write

$$\tau = 2p_{26} + 3p_{46}, \quad -\sigma = 3p_{24} + 2p_{44}, \quad \gamma = p_{27}.$$

Expanding the Euler–Jacobian determinant in the standard-monomial basis (13), the coefficients at q^4, q^3y, q^2y^2 are

$$E_0 = 2p_{26}p_{27} + 3p_{27}p_{46} + 3 = 3 + \gamma\tau, \quad (19)$$

$$E_1 = 3p_{24}p_{27} - 2p_{26} + 2p_{27}p_{44} - 3p_{46} = -(\tau + \gamma\sigma), \quad (20)$$

$$E_2 = 6p_{24} + 4p_{26}^2 + 12p_{26}p_{46} + 4p_{44} + 9p_{46}^2 = \tau^2 - 2\sigma. \quad (21)$$

These are actual generators of the universal mixed-partial, Euler, determinant, and raw-Pluecker coefficient ideal.

Lemma 7.4 (Newton identity with constant 3). *The rows above satisfy*

$$\gamma^2 E_2 - (\gamma\tau - 1)E_0 - 2\gamma E_1 = 3. \quad (22)$$

Proof. Substitute $E_0 = 3 + \gamma\tau$, $E_1 = -(\tau + \gamma\sigma)$, and $E_2 = \tau^2 - 2\sigma$ and expand. Equivalently, extend M by zero on the augmentation summand and apply the degree-three Newton identities to $I_3 + \gamma\widetilde{M}$. $\square$

Remark 7.5. The constants 2 and 3 arise in the Euler first column itself. The final contradiction uses membership of the displayed coefficient rows in the Keller ideal; it does not use a trace of a chain boundary.

8. PRIMEWISE COMPANION BINDING

Let R be the unsaturated coefficient ring of the stationary package and let $J_K \subset R$ be the full Keller coefficient ideal. It is useful to write

$$J_K = J_{\text{alg}} + J_{\text{mix}} + J_E + J_{\text{det}} + J_{\text{Pl}},$$

where the five summands are generated by the algebra/multiplication equations, mixed partials, Euler first-column equations, determinant coefficients, and raw Pluecker relations, respectively.

Lemma 8.1 (Primewise exclusion). *It is enough to prove $J_K R_{\mathfrak{p}} = R_{\mathfrak{p}}$ at the generic prime $\mathfrak{p}$ of every pointed companion component.*

Proof. Every irreducible component has a generic prime containing J_K . If the localized ideal is the unit ideal, such a prime cannot exist. Treating generic primes separately retains every coefficient-divisor branch. $\square$

For $n = 2, 3$, let U_n be the coordinate ring representing the universal pointed rank- n package: its variables are the pointed algebra coefficients, the allowed ordinary-degree source slots, the graph-output slots, the Euler slots, and the determinant and raw Pluecker slots. Let

$$J_n^{\text{univ}} = J_{n,\text{alg}} + J_{n,\text{mix}} + J_{n,E} + J_{n,\text{det}} + J_{n,\text{Pl}}$$

be its universal Keller ideal. By (17) and (19)–(21),

$$Q_0, Q_1 \in J_2^{\text{univ}}, \quad E_0, E_1, E_2 \in J_3^{\text{univ}}.$$

Theorem 8.2 (Primewise support-compatible binding). *Let $\mathfrak{p}$ be the generic prime of a pointed rank- n companion component supplied by Theorem 4.4, and put $A = R_{\mathfrak{p}}$. There is a polynomial specialization*

$$\varphi_{\mathfrak{p}} : U_n \longrightarrow A, \quad n = 2 \text{ or } 3,$$

which carries the universal companion diagram to the selected companion direction and satisfies

$$\varphi_{\mathfrak{p}}(J_n^{\text{univ}}) \subset J_K A.$$

Every other simultaneous key remains in A as a coefficient variable.

Proof. The pointed normal form gives the universal basis $1, W, \dots, W^{n-1}$. Send the universal structure coefficients, all allowed source slots, and all retained graph-output variables to their actual values at $\mathfrak{p}$. All other simultaneous keys and relations among them remain in the coefficient algebra; they are neither differentiated away nor diagonalized.

Monic division in the selected key is polynomial. Consequently:

- (a) the universal multiplication equations map to the actual multiplication relations of the selected pointed algebra;
- (b) the universal mixed-partial rows map to the actual mixed-partial graph relations because the relevant derivative outputs were retained in the stationary diagram;
- (c) the Euler first column maps coefficientwise to the actual Euler column;
- (d) determinant coefficients map to the corresponding actual determinant coefficients, since every product space entering the homogeneous determinant was retained before specialization;
- (e) raw Pluecker generators map to the actual raw minors, before any saturation.

Thus each summand of J_n^{univ} maps into the corresponding summand of $J_K A$. The finite unsaturated package is formed before localization; localization is used only to work at the generic prime, not to delete a coefficient-divisor branch. $\square$

Theorem 8.3 (Exclusion of pointed companions). *No pointed quadratic or cubic companion component occurs in the first-disappearance package of a genuine degree-six Keller map.*

Proof. At a quadratic generic prime, Theorem 8.2 makes Q_0, Q_1 zero modulo the actual localized Keller ideal, while (18) gives $2 = 0$. At a cubic generic prime it makes E_0, E_1, E_2 zero, while (22) gives $3 = 0$. Both are impossible in characteristic zero. Apply Theorem 8.1. $\square$

Remark 8.4 (The local fold calibration). The algebra $A[H]/(H^2 - z)$ and the map

$$(H, y, c) \mapsto \left(y, c, -\frac{H^2}{2} \right)$$

have Jacobian $-H$. Thus a bare monic algebra and a leading fold equation are insufficient. The preceding theorem uses the full universal coefficient ideal; the fold does not satisfy the terminal Euler rows.

9. COMPLETION OF THE PROOF

Proof of Theorem 1.1. Assume that F is nonproper. Choose E by Theorem 2.2, run the tuned relative MMP to first disappearance, and form $\mathcal{K}_v^{\leq 6}$.

Apply Theorem 4.4 at the generic point of every irreducible first-disappearance branch. A regular point branch is excluded by Theorem 5.6; a regular curve branch is excluded by Theorem 5.8. A nonregular branch contains a pointed rank-two or rank-three companion, excluded by Theorem 8.3. The alternatives are componentwise exhaustive and retain all simultaneous relations and coefficient-divisor branches.

Thus the selected negative-defect divisor has no possible first disappearance, contradicting Theorem 2.3. Hence F is proper. A proper étale endomorphism of $\mathbb{A}_{\mathbb{C}}^3$ is finite étale; affine space has no nontrivial connected finite étale cover, so F has degree one and is a polynomial automorphism. $\square$

10. DEGREE-SEVEN CALIBRATION

With $u = 1 + xy$, put

$$\begin{aligned} f_1 &= u^3 z + y^2 u(4 + 3xy), \\ f_2 &= y + 3xu^2 z + 3xy^2(4 + 3xy), \\ f_3 &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

A direct differentiation gives

$$\det J(f_1, f_2, f_3) = -2, \quad (\deg f_1, \deg f_2, \deg f_3) = (7, 6, 4),$$

and the three points

$$(0, 0, -1/4), \quad (1, -3/2, 13/2), \quad (-1, 3/2, 13/2)$$

have the same image $(-1/4, 0, 0)$. In the hyperbolic signature $(1, 2)$, the descended slot $u^3 v$ has ordinary cost seven and is absent from the degree-six support. Thus the degree-six argument is sharply calibrated at the first known three-dimensional counterexample.

APPENDIX A. THE DIRECT $(1, 2, 3)$ COEFFICIENT EQUATIONS

The polynomial Wronski syzygy. Let C_μ denote the coefficient of $h^{\mu_0}x^{\mu_1}y^{\mu_2}$ in $\mathcal{W}(F, p) - h^{15}$. The nonzero multipliers in the syzygy used in Theorem 6.2 are the following.

coefficient monomial	multiplier λ_μ
h^{15}	$-B^5$
$h^{13}x$	$B^4(A+2)$
$h^{13}y$	$-B^5$
$h^{11}x^2$	$-B^3(A+2)^2$
$h^{11}xy$	$B^4(A+2)$
$h^{11}y^2$	$-B^5$
h^9x^3	$B^2(A+2)^3$
h^9x^2y	$-B^3(A+2)^2$
h^9xy^2	$B^4(A+2)$
h^9y^3	$-B^5$
h^7x^4	$-B(A+2)^4$
h^7x^3y	$B^2(A+2)^3$
$h^7x^2y^2$	$-B^3(A+2)^2$
h^7xy^3	$B^4(A+2)$
h^7y^4	$-B^5$
h^5x^5	$(A+2)^5$
h^5x^4y	$-B(A+2)^4$
$h^5x^3y^2$	$B^2(A+2)^3$
$h^5x^2y^3$	$-B^3(A+2)^2$
h^5xy^4	$B^4(A+2)$
h^5y^5	$-B^5$
h^3x^5y	$5A^4 + 30A^3 + 70A^2 + 75A + 31$
$h^3x^4y^2$	$-B(2A+3)(2A^2+6A+5)$
$h^3x^3y^3$	$B^2(3A^2+9A+7)$
$h^3x^2y^4$	$-B^3(2A+3)$
h^3xy^5	B^4

A direct expansion gives

$$\sum_{\mu} \lambda_{\mu}[p_{ij}]C_{\mu} = 0 \quad \text{for every } p_{ij}, \quad \sum_{\mu} \lambda_{\mu}C_{\mu} = -B^5.$$

The closed Pluecker fibre. After setting $B = 0$, imposing $p_{78} = 1$, and substituting (8), the coefficient equations are generated, after clearing nonzero rational denominators, by the following polynomials.

$$\begin{aligned}
R_0 &= 2a_6 - b_5, \\
R_1 &= -2A + a_5 - 2b_4, \\
R_2 &= Ab_6 + 3a_3 - 2a_5b_6 + 2a_6b_5 \\
&\quad - b_2, \\
R_3 &= 4Aa_6 - Ab_5 + 2a_2 - 4a_4b_6 \\
&\quad + 4a_6b_4 - 2b_1, \\
R_4 &= 2Aa_5 - 3Ab_4 + 3A + a_1
\end{aligned}$$

$$\begin{aligned}
& -2a_4b_5 + 2a_5b_4 - 3b_0 + 3, \\
R_5 &= -2Ab_3 + 2a_2b_6 - 3a_3b_5 + 3a_5b_3 \\
& - 2a_6b_2, \\
R_6 &= -6Aa_3 + 3Aa_5b_6 - 3Aa_6b_5 + 2Ab_6 \\
& + 4a_1b_6 - a_2b_5 - 6a_3b_4 + 6a_4b_3 \\
& + a_5b_2 - 4a_6b_1 + 2b_6, \\
R_7 &= -4Aa_2 + 6Aa_4b_6 - 6Aa_6b_4 + 6Aa_6 \\
& + 2Ab_1 - Ab_5 + 6a_0b_6 + a_1b_5 \\
& - 4a_2b_4 + 4a_4b_2 - a_5b_1 - 6a_6b_0 \\
& + 6a_6 - b_5, \\
R_8 &= -2Aa_1 + 3Aa_4b_5 - 3Aa_5b_4 + 3Aa_5 \\
& + 4Ab_0 - 4Ab_4 + 3a_0b_5 - 2a_1b_4 \\
& + 2a_4b_1 - 3a_5b_0 + 3a_5 - 4b_4, \\
R_9 &= -Aa_3b_6 + Aa_6b_3 + 3a_2b_3 - 3a_3b_2, \\
R_{10} &= 2Aa_2b_6 - 5Aa_3b_5 + 5Aa_5b_3 - 2Aa_6b_2 \\
& + 4Ab_3 + 6a_1b_3 - 6a_3b_1 + 4b_3, \\
R_{11} &= 5Aa_1b_6 - 2Aa_2b_5 - 9Aa_3b_4 + 9Aa_3 \\
& + 9Aa_4b_3 + 2Aa_5b_2 - 4Aa_5b_6 - 5Aa_6b_1 \\
& + 4Aa_6b_5 + Ab_2 + 9a_0b_3 + 3a_1b_2 \\
& - 3a_2b_1 - 9a_3b_0 + 9a_3 - 4a_5b_6 \\
& + 4a_6b_5 + b_2, \\
R_{12} &= 8Aa_0b_6 + Aa_1b_5 - 6Aa_2b_4 + 6Aa_2 \\
& + 6Aa_4b_2 - 8Aa_4b_6 - Aa_5b_1 - 8Aa_6b_0 \\
& + 8Aa_6b_4 - 2Ab_1 + 6a_0b_2 - 6a_2b_0 \\
& + 6a_2 - 8a_4b_6 + 8a_6b_4 - 2b_1, \\
R_{13} &= 4Aa_0b_5 - 3Aa_1b_4 + 3Aa_1 + 3Aa_4b_1 \\
& - 4Aa_4b_5 - 4Aa_5b_0 + 4Aa_5b_4 - 5Ab_0 \\
& + 3a_0b_1 - 3a_1b_0 + 3a_1 - 4a_4b_5 \\
& + 4a_5b_4 - 5b_0, \\
R_{14} &= 2Aa_2b_3 - 2Aa_3b_2 + Aa_3b_6 - Aa_6b_3 \\
& + a_3b_6 - a_6b_3, \\
R_{15} &= 8Aa_1b_3 - 2Aa_2b_6 - 8Aa_3b_1 + 7Aa_3b_5 \\
& - 7Aa_5b_3 + 2Aa_6b_2 - 2a_2b_6 + 7a_3b_5 \\
& - 7a_5b_3 + 2a_6b_2, \\
R_{16} &= 12Aa_0b_3 + 4Aa_1b_2 - 6Aa_1b_6 - 4Aa_2b_1 \\
& + 3Aa_2b_5 - 12Aa_3b_0 + 12Aa_3b_4 - 12Aa_4b_3 \\
& - 3Aa_5b_2 + 6Aa_6b_1 - 6a_1b_6 + 3a_2b_5 \\
& + 12a_3b_4 - 12a_4b_3 - 3a_5b_2 + 6a_6b_1, \\
R_{17} &= 8Aa_0b_2 - 10Aa_0b_6 - Aa_1b_5 - 8Aa_2b_0 \\
& + 8Aa_2b_4 - 8Aa_4b_2 + Aa_5b_1 + 10Aa_6b_0 \\
& - 10a_0b_6 - a_1b_5 + 8a_2b_4 - 8a_4b_2 \\
& + a_5b_1 + 10a_6b_0, \\
R_{18} &= 4Aa_0b_1 - 5Aa_0b_5 - 4Aa_1b_0 + 4Aa_1b_4
\end{aligned}$$

$$\begin{aligned}
& -4Aa_4b_1 + 5Aa_5b_0 - 5a_0b_5 + 4a_1b_4 \\
& -4a_4b_1 + 5a_5b_0, \\
R_{19} &= (A+1)(a_2b_3 - a_3b_2), \\
R_{20} &= (A+1)(a_1b_3 - a_3b_1), \\
R_{21} &= (A+1)(3a_0b_3 + a_1b_2 - a_2b_1 - 3a_3b_0), \\
R_{22} &= (A+1)(a_0b_2 - a_2b_0), \\
R_{23} &= (A+1)(a_0b_1 - a_1b_0).
\end{aligned}$$

The final five rows factor as $A+1$ times the 2×2 minors of the two truncated coefficient vectors

$$(a_0, a_1, a_2, a_3), \quad (b_0, b_1, b_2, b_3).$$

They therefore separate the special value $A = -1$ from the rank-one branch in which these vectors are proportional. The first nineteen equations retain the higher coefficients $a_4, a_5, a_6, b_4, b_5, b_6$ and eliminate both branches simultaneously.

Thus Theorem 6.3 is the finite assertion

$$(R_0, \dots, R_{23}) = (1) \quad \text{in} \quad \mathbb{Q}[A, a_0, \dots, a_6, b_0, \dots, b_6].$$

APPENDIX B. THE FORTY REGULAR NORMAL FORMS

Write

$$A = \sum_{(i,j) \in S_A} a_{ij} u^i v^j, \quad B = \sum_{(i,j) \in S_B} b_{ij} u^i v^j, \quad \Lambda = 1 + \sum_{(i,j) \in S_\Lambda \setminus \{(0,0)\}} \ell_{ij} u^i v^j,$$

with the supports in (7). For $r < s$, the linear parts are normalized to $A_1 = v$ and $B_1 = u$. Each row below prescribes the additional normalizations and the single open coefficient, if any. On an open row the ideal I_ν also contains $z_f f - 1$; a row marked “closed” has no Rabinowitsch variable. The final two columns record the number of variables and coefficient equations after the substitutions.

No.	(r, s)	normalization	open
1	(2, 4)	$\ell_{01} = 0, \ell_{10} = 1$	closed
2	(2, 4)	$\ell_{01} = 1$	closed
3	(2, 3)	$\ell_{01} = 0, \ell_{10} = 1$	closed
4	(2, 3)	$\ell_{01} = 1$	closed
5	(1, 3)	$\ell_{01} = 1$	closed
6	(1, 3)	$\ell_{01} = 0, \ell_{20} = 1$	closed
7	(1, 3)	$\ell_{01} = 0, \ell_{10} = 1, \ell_{20} = 0$	closed
8	(1, 4)	$\ell_{01} = 1$	closed
9	(1, 4)	$\ell_{01} = 0, \ell_{20} = 1$	closed
10	(1, 4)	$\ell_{01} = 0, \ell_{10} = 1, \ell_{20} = 0$	closed
11	(1, 1)	$\ell_{01} = 0, \ell_{02} = 0, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	closed
12	(1, 1)	$\ell_{01} = 0, \ell_{02} = 0, \ell_{11} = 0, \ell_{20} = 1$	closed
13	(1, 1)	$\ell_{01} = 1, \ell_{02} = 0, \ell_{10} = 0, \ell_{11} = 0, \ell_{20} = 1$	closed
14	(1, 1)	$\ell_{01} = 0, \ell_{02} = 0, \ell_{10} = 0, \ell_{11} = 1, \ell_{20} = 0$	closed
15	(1, 1)	$\ell_{02} = 0, \ell_{10} = 1, \ell_{11} = 1, \ell_{20} = 0$	closed
16	(1, 2)	$\ell_{01} = 0, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	closed
17	(1, 2)	$\ell_{01} = 1, \ell_{10} = 0, \ell_{11} = 0, \ell_{20} = 0$	closed
18	(1, 2)	$b_{21} = 0, b_{30} = 0, \ell_{01} = 1, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	$a_{40} \neq 0$
19	(1, 2)	$a_{40} = 0, b_{02} = 0, b_{21} = 0, \ell_{01} = 1, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	$a_{12} \neq 0$
20	(1, 2)	$a_{12} = 0, a_{21} = 0, a_{40} = 0, b_{21} = 0, b_{30} = 0, \ell_{01} = 1, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	$a_{30} \neq 0$
21	(1, 2)	$a_{12} = 0, a_{21} = 0, a_{30} = 0, a_{40} = 0, \ell_{01} = 1, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	closed
22	(1, 2)	$a_{12} = 0, a_{40} = 0, b_{21} = 0, b_{30} = 0, \ell_{01} = 1, \ell_{10} = 1, \ell_{11} = 0, \ell_{20} = 0$	$a_{21} \neq 0$
23	(1, 2)	$b_{21} = 0, b_{30} = 0, \ell_{11} = 1, \ell_{20} = 0$	$a_{40} \neq 0$
24	(1, 2)	$a_{40} = 0, b_{02} = 0, b_{21} = 0, \ell_{11} = 1, \ell_{20} = 0$	$a_{12} \neq 0$
25	(1, 2)	$a_{12} = 0, a_{40} = 0, \ell_{11} = 1, \ell_{20} = 0$	closed
26	(1, 2)	$b_{21} = 0, b_{30} = 0, \ell_{11} = 1, \ell_{20} = 1$	$a_{40} \neq 0$
27	(1, 2)	$a_{40} = 0, b_{02} = 0, b_{21} = 0, \ell_{11} = 1, \ell_{20} = 1$	$a_{12} \neq 0$
28	(1, 2)	$a_{12} = 0, a_{21} = 0, a_{40} = 0, b_{21} = 0, b_{30} = 0, \ell_{11} = 1, \ell_{20} = 1$	$a_{30} \neq 0$
29	(1, 2)	$a_{12} = 0, a_{21} = 0, a_{30} = 0, a_{40} = 0, \ell_{11} = 1, \ell_{20} = 1$	closed
30	(1, 2)	$a_{12} = 0, a_{40} = 0, b_{21} = 0, b_{30} = 0, \ell_{11} = 1, \ell_{20} = 1$	$a_{21} \neq 0$
31	(1, 2)	$b_{21} = 0, \ell_{01} = 0, \ell_{11} = 0, \ell_{20} = 1$	$a_{40} \neq 0$
32	(1, 2)	$a_{40} = 0, b_{02} = 0, b_{21} = 0, b_{30} = 0, \ell_{01} = 0, \ell_{11} = 0, \ell_{20} = 1$	$a_{12} \neq 0$
33	(1, 2)	$a_{12} = 0, a_{40} = 0, b_{21} = 0, b_{30} = 0, \ell_{01} = 0, \ell_{11} = 0, \ell_{20} = 1$	$a_{21} \neq 0$
34	(1, 2)	$a_{12} = 0, a_{21} = 0, a_{40} = 0, b_{21} = 0, \ell_{01} = 0, \ell_{11} = 0, \ell_{20} = 1$	$a_{30} \neq 0$
35	(1, 2)	$a_{12} = 0, a_{21} = 0, a_{30} = 0, a_{40} = 0, \ell_{01} = 0, \ell_{11} = 0, \ell_{20} = 1$	closed
36	(1, 2)	$b_{21} = 0, \ell_{01} = 1, \ell_{11} = 0, \ell_{20} = 1$	$a_{40} \neq 0$

No.	(r, s)	normalization	open
37	$(1, 2)$	$a_{40} = 0, b_{02} = 0, b_{21} = 0, b_{30} = 0, \ell_{01} = 1, \ell_{11} = 0, \ell_{20} = 1$	$a_{12} \neq 0$
38	$(1, 2)$	$a_{12} = 0, a_{40} = 0, b_{21} = 0, b_{30} = 0, \ell_{01} = 1, \ell_{11} = 0, \ell_{20} = 1$	$a_{21} \neq 0$
39	$(1, 2)$	$a_{12} = 0, a_{21} = 0, a_{40} = 0, b_{21} = 0, \ell_{01} = 1, \ell_{11} = 0, \ell_{20} = 1$	$a_{30} \neq 0$
40	$(1, 2)$	$a_{12} = 0, a_{21} = 0, a_{30} = 0, a_{40} = 0, \ell_{01} = 1, \ell_{11} = 0, \ell_{20} = 1$	closed

The first fifteen rows give the orbit counts

(r, s)	$(2, 4)$	$(2, 3)$	$(1, 3)$	$(1, 4)$	$(1, 1)$
# forms	2	2	3	3	5.

Rows 16–40 are the five families for $(1, 2)$, subdivided by the first nonzero coefficient among $a_{40}, a_{12}, a_{21}, a_{30}$ and the terminal closed branch. Hence the total number of locally closed forms is

$$2 + 2 + 3 + 3 + 5 + 25 = 40.$$

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